Closed loop control of three-phase AC-DC converter-II

Introduction

In the realm of power electronics, the efficient and reliable conversion of AC to DC power is paramount, especially in applications like electric vehicle charging infrastructure. This lecture delves into the sophisticated techniques required for the closed-loop control of three-phase AC-DC converters. The primary objectives are twofold: first, to precisely regulate the output DC voltage to a desired value, ensuring stable power delivery; and second, to achieve a unity power factor at the AC input, meaning the current drawn from the source is perfectly in phase with the AC voltage. This unity power factor operation is crucial for minimizing reactive power, reducing losses, and improving the overall quality of power drawn from the grid.

Achieving these objectives in a three-phase system presents unique challenges. The inherent coupling between phases in a balanced three-phase system (where the sum of instantaneous currents Isa + Isb + Isc equals zero) prevents independent control of individual phase currents. This limitation necessitates a transformation of the system's representation into a domain where control can be exercised independently. This lecture explores the use of space vectors and different reference frames—specifically the stationary alpha-beta frame and the rotating dq frame—to decouple these variables, thereby enabling precise and independent control over the converter's operation.

Core Concepts

Term	Definition	Significance
`Space Vector`	A rotating vector in a two-dimensional spatial domain that represents the combined effect of three-phase quantities (voltages or currents). Its magnitude and angle vary over time.	Simplifies the representation of three-phase systems into a single vector, making analysis and control more intuitive and efficient.
	A stationary, two-dimensional coordinate system (alpha and	Provides a decoupled representation of three-phase

Term	Definition	Significance
`Alpha-Beta Reference Frame`	beta axes 90 degrees apart) onto which the three-phase space vector is projected.	quantities in a stationary frame, allowing for independent control of alpha and beta components.
`DQ Reference Frame`	A rotating, two-dimensional coordinate system (direct 'd' and quadrature 'q' axes 90 degrees apart) that rotates synchronously with the space vector, often aligned with the supply voltage vector.	Decouples the control variables into DC quantities under steady-state conditions, simplifying controller design and enabling independent control of active and reactive power components.
`Synchronous Rotating Frame`	Another name for the DQ reference frame, emphasizing its rotation at the same angular speed as the AC system's fundamental frequency.	Highlights the frame's ability to transform AC quantities into DC quantities, which are easier to regulate with conventional PI controllers.
`Cross-Coupling Terms`	Terms in the DQ-axis models that link the d-axis and q-axis components, typically involving the angular frequency (ω) and inductance (L).	These terms indicate interaction between the d and q axes, which must be compensated for (e.g., via feedforward control) to achieve true independent control.
`Independent Control`	The ability to adjust one control variable (e.g., d-axis current) without affecting another (e.g., q-axis current).	Essential for achieving precise control objectives like unity power factor (by setting q-axis current to zero) and output voltage regulation (via d-axis current).
`Winding Resistance (Rs)`	The inherent series resistance of the inductor windings in the converter's input filter.	Represents a real-world loss component in the converter model, affecting efficiency

Term	Definition	Significance
		and requiring consideration in the control design.

Detailed Analysis

1. Control Objectives and the Challenge of Three-Phase Coupling

The primary goals for a three-phase AC-DC converter are to regulate the output DC voltage to a desired setpoint and to ensure the current drawn from the AC source maintains a unity power factor. Unity power factor implies that the input current is perfectly in phase with the AC voltage, minimizing reactive power exchange with the grid. However, a fundamental challenge arises in balanced three-phase systems: the instantaneous sum of the three-phase currents (Isa + Isb + Isc) is always zero. This constraint means that the three phase currents are not truly independent; changing one phase current automatically affects the others. This inherent coupling makes direct independent control of each phase current difficult, necessitating a transformation to a different coordinate system.

2. Space Vectors and Reference Frame Transformations

To overcome the coupling issue, the concept of a space vector is introduced. A space vector is a single vector rotating in a two-dimensional plane, representing the combined effect of the three-phase quantities (voltages or currents). This vector rotates at the same frequency as the AC system. To enable independent control, these space vectors are projected onto different reference frames:

- **Alpha-Beta Stationary Reference Frame**: This is a fixed, two-dimensional coordinate system where the alpha-axis is typically aligned with the peak of one of the phase voltages (e.g., phase A). The beta-axis is 90 degrees apart. In this frame, the components of the space vector (Vs_alpha, Vs_beta) are independent of each other. For example, Vs_alpha = (3/2) * Vsa, and Vs_beta = (sqrt(3)/2) * (Vsb - Vsc). Similar transformations apply to currents (Is_alpha, Is_beta) and converter voltages (V_converter_alpha, V_converter_beta).
- **DQ Rotating Reference Frame**: This frame rotates synchronously with the space vector, specifically aligned with the supply voltage space vector (Vs). The d-axis (direct axis) is aligned with Vs, and the q-axis (quadrature axis) is 90 degrees ahead. The key advantage of aligning the d-axis with Vs is that the q-component of the supply voltage (Vsq) becomes zero, simplifying the control. The transformation from alpha-beta to dq involves multiplying by

`e^(-jtheta)`, where `theta` is the angle of the rotating frame. This transformation converts AC quantities into DC quantities under steady-state conditions, making them easier to control with conventional PI controllers.

3. Single-Phase Equivalent Circuit and Space Vector Representation

Before deriving the DQ model, a single-phase equivalent circuit is considered. This circuit includes the supply voltage (Vs), an inductor (L), a series winding resistance (Rs) of the inductor, and the voltage applied by the converter (V_converter). The current flowing is Is.
When extended to space vectors, the system can be represented as:
Vs_space_vector = L * d/dt (Is_space_vector) + Rs * Is_space_vector + V_converter_space_vector.
Here, Vs_space_vector, Is_space_vector, and V_converter_space_vector represent the combined effect of the three-phase quantities. The d-axis is aligned with the Vs_space_vector, meaning Vsd is non-zero, but Vsq is zero.

4. Derivation of the DQ Axis Model

The derivation of the DQ axis model starts from the KVL equation in the stationary alpha-beta frame and transforms it into the rotating DQ frame.
The KVL equation in the stationary frame is:
`Vs_space_vector = L * d/dt (Is_space_vector) + Rs * Is_space_vector + V_converter_space_vector`

To transform this into the DQ frame, each space vector is multiplied by `e^(-jtheta)`, where `theta` is the angle of the rotating frame (`theta = omega*t`). This effectively shifts the reference frame.
The differentiation term `L * d/dt (Is_space_vector * e^(-jtheta))` is crucial. Since both `Is_space_vector` and `e^(-jtheta)` are time-varying, the product rule for differentiation must be applied. This leads to:
`L * d/dt (Is_space_vector_dq) + j * omega * L * Is_space_vector_dq`
where `Is_space_vector_dq` is the current space vector in the DQ frame.

After applying this transformation and separating the real (d-axis) and imaginary (q-axis) components, the following independent DQ-axis models are obtained:

D-axis Model:
`L * d/dt (Isd) + Rs * Isd = Vsd - V_converter_D + omega * L * Isq`

Q-axis Model:
`L * d/dt (Isq) + Rs * Isq = Vsq - V_converter_Q - omega * L * Isd`

In these equations:
* `Isd` and `Isq` are the d-axis and q-axis components of the current.

* `Vsd` and `Vsq` are the d-axis and q-axis components of the supply voltage. Since the d-axis is aligned with Vs, `Vsq = 0`.
* `V_converter_D` and `V_converter_Q` are the d-axis and q-axis components of the converter voltage, which are the control inputs.
* `Rs` is the series winding resistance.
* `L` is the inductance.
* `omega` is the angular frequency (2 * pi * fs).

5. Independent Control and Cross-Coupling Terms

A critical observation from the derived DQ models is the presence of **cross-coupling terms**: `omega * L * Isq` in the d-axis equation and `-omega * L * Isd` in the q-axis equation. These terms indicate that the d-axis current dynamics are influenced by the q-axis current, and vice-versa. While the DQ transformation provides a framework for independent control, these cross-coupling terms must be actively compensated for (e.g., using feedforward control) to achieve true decoupling.

Once these terms are managed, the d-axis and q-axis current components can be controlled independently. This is a significant advantage because:
* **Unity Power Factor Control**: To achieve unity power factor, the reactive current component (Isq) needs to be zero. By independently controlling Isq, it can be driven to zero.
* **Output Voltage Regulation**: The active current component (Isd) is directly related to the power flow and thus the output DC voltage. By independently controlling Isd, the output voltage can be regulated.

The ability to control Isd and Isq independently in the DQ frame directly translates to independent control over the active and reactive power drawn from the three-phase AC source, which then reflects in the balanced three-phase quantities. This independent control is fundamental to achieving the desired performance objectives of the AC-DC converter.

Key Takeaways

- [ ] Three-phase systems require transformation to a different reference frame for independent control of currents.
- [ ] Space vectors simplify the representation of three-phase quantities into a single rotating vector.
- [ ] The alpha-beta stationary frame provides a decoupled 2D representation of three-phase quantities.
- [ ] The DQ rotating frame (synchronous rotating frame) transforms AC quantities into DC quantities under steady-state, simplifying control.

- [ ] Aligning the d-axis with the supply voltage space vector (Vs) makes the q-component of the supply voltage (Vsq) zero.
- [ ] The derivation of the DQ-axis model involves applying KVL in the stationary frame and then transforming it to the rotating frame, leading to cross-coupling terms.
- [ ] Cross-coupling terms (e.g., `omega * L * Isq`) link the d-axis and q-axis dynamics and must be compensated for to achieve true independent control.
- [ ] Independent control of Isd and Isq in the DQ frame enables precise regulation of output voltage and achievement of unity power factor.

Conclusion

The detailed analysis of space vectors and reference frame transformations, particularly the DQ rotating frame, lays the essential groundwork for advanced closed-loop control of three-phase AC-DC converters. By transforming the inherently coupled three-phase system into a decoupled two-axis (d and q) model, engineers gain the ability to independently control critical parameters like active and reactive current components. This independent control is not merely an academic exercise; it is the cornerstone for achieving high-performance power conversion, enabling precise output voltage regulation and the crucial unity power factor operation that minimizes grid impact and maximizes energy efficiency. While cross-coupling terms present a challenge, their identification within the DQ model allows for targeted compensation strategies, paving the way for robust and reliable converter operation in real-world applications, from industrial power supplies to renewable energy integration and electric vehicle charging. The insights gained from this derivation are fundamental to designing effective control algorithms that will be explored further in subsequent discussions.